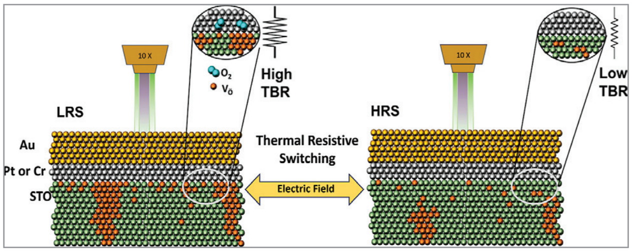
A revolutionary method found for heat management in devices

Researchers at CiQUS have uncovered a method to modulate device thermal resistance using voltage. Published in ACS Applied Materials & Interfaces and led by Rafael Ramos and Francisco Rivadulla, the study targets materials

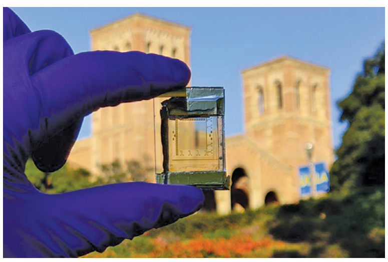


Memristor (Credit: https://www.usc.es/ciqus)

with adjustable thermal conductivity, addressing heat dissipation in electronics, notably memristors. These components exhibit resistive switching, altering electrical resistance via an electric field. The team found this switching triggers a thermal resistive effect at metal-oxide interfaces, modifiable by about 20% at room temperature. Part of the MEMTHERM project, aiming at ion movement control in dielectric oxides, this discovery could revolutionise heat management in various technologies, from mobile phones to processors, enhancing performance and sustainability.

A compact device that enhances visual computing

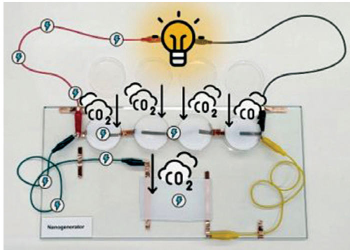
A study from the California NanoSystems Institute at UCLA has introduced a new device that enhances visual computing. This compact device features a small array of transparent pixels on a 1cm square plane made of a thin 2D semiconductor, liquid crystal, and electrodes. It rapidly and efficiently changes transparency with minimal light, improving image processing by directly manipulating images without converting them to digital signals. Integrated into smartphone cameras, it reduces glare and speeds up processing, reducing data sent to the cloud. This advancement offers potential applications across enhanced vehicle sensing to advanced image encryption, making high-resolution imaging and sensing more accessible and efficient.



This experimental device uses a 2D semiconductor material developed by Xiangfeng Duan, UCLA professor of chemistry and biochemistry (Courtesy of Dehui Zhang)

Carbon-dioxide based nanogenerator for electricity

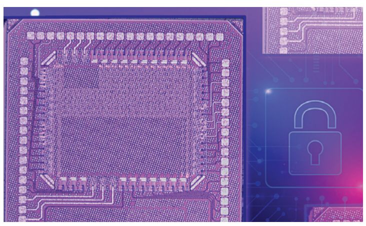
Researchers at the University of Queensland, led by Dr Zhuyuan Wang, have created a carbon-negative generator that converts carbon-dioxide (CO2) into electricity. This nanogenerator, made from a polyamine gel and a boron nitrate skeleton, produces ions, generating a diffusion current for electricity. Operating on a principle like natural processes in the human body, it uses ion movement for efficient energy conversion. Tests in CO2-rich environments confirmed its effectiveness. Harnessing about 1% of CO2 gas's energy, the technology holds promise for personal devices and integration with industrial CO2 capture for electricity generation. Part of the GETCO2 initiative, it aims to redefine CO2 as a resource for sustainable energy.



The UQ-developed technology absorbs carbon-dioxide and generates electricity (Credit: https://www.uq.edu.au)

New ML chip provides robust security against common attack methods

Researchers from MIT and the MIT-IBM Watson AI Lab have developed a new type of machine-learning accelerator that provides robust security against the two most common attack methods, protecting sensitive data while maintaining performance. This accelerator is designed to ensure high security with minimal effects on speed and accuracy, and is suitable for AI-driven applications such as augmented reality and autonomous driving. Although implementing these secure chips slightly increases costs and reduces energy efficiency, these trade-offs are generally acceptable for the added security. The accelerator uses a new digital in-memory compute chip that integrates security measures into the hardware, performs computations within the device's memory, and minimises external data transfers, which enhances processing speed and security against side-channel and bus-probing attacks. The chip encrypts data and generates unique decryption keys internally, effectively preventing breaches.



A new chip can efficiently accelerate machine-learning workloads on edge devices like smartphones while protecting sensitive user data from two common types of attacks—side-channel and bus-probing (Credits: Chip figure courtesy of the researchers; MIT News; iStock)